

FORM 2
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COMPLETE SPECIFICATION
(See section 10 and M.13)

Tumbler Holder for RO Purifier

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The following specification particularly describes the invention and the manner in which it is to be performed.

EXTENSIBLE TUMBLER HOLDER FOR RO PURIFIER

FIELD OF INVENTION

The present invention relates to the field of household appliances, specifically to accessory attachments for water purification systems. More particularly, it relates to an extensible tumbler holder designed for reverse osmosis (RO) purifiers, enhancing user convenience by providing adjustable storage solutions for varying sizes of tumblers.

BACKGROUND OF INVENTION

Obtaining clean, potable water is a fundamental necessity for maintaining a good quality of life. However, the task of collecting water from a wall-mounted reverse osmosis purifier can present unique challenges that hinder this basic requirement. This invention aims to investigate the difficulties encountered in the process of filling a tumbler with water from a heightened wall-mounted reverse osmosis purifier, and to explore potential solutions to address these challenges.

The use of reverse osmosis technology in water purification has gained widespread popularity due to its effectiveness in removing a wide range of contaminants, including dissolved salts, bacteria, and other impurities.

While extensive research has been conducted on the application of reverse osmosis in remote communities, the practical implementation of these technologies often faces significant obstacles that have limited their widespread adoption (Rabindra, 2018). One such challenge is the difficulty in accessing the water from a wall-mounted purifier, particularly when the height of the dispenser is elevated.

SUMMARY OF INVENTION

Evaluating the Specifications of a RO Purifier Stand:

The importance of adaptable water treatment systems has been highlighted in prior research. Flexible models for simulating water treatment stations have been developed, emphasizing the ease and flexibility of implementation for a wide range of purification plants. (Ahmed & Abdelrahman, 2009) Additionally, studies have explored the optimization of reverse osmosis module networks, considering factors such as module structure, system configuration, and operating conditions to maximize efficiency and profitability. (Maskan et al., 2000)

The reverse osmosis (Maskan et al., 2000) purifier under investigation is 46cm in length and is mounted on a stand that can be adjusted to a height of 96cm, double the size of the RO unit. The horizontal length of the RO purifier is 25cm, and the tray can be pulled out to a length of 50cm, allowing for the placement of tumblers of varying sizes. This adaptable design provides flexibility and convenience for users, catering to diverse water purification needs.

DETAIL DESCRIPTION

The holder is constructed from durable materials, designed to withstand regular use. A sliding mechanism allows the length of the holder to be adjusted easily, ensuring it can hold tumblers of different sizes securely. Users can pull out the holder's width for easy access and storage of multiple tumblers at once.

The tumbler holder is mounted securely on the back of an RO purifier, ensuring it does not interfere with the purifier's operation or aesthetics. Made from high-quality, durable materials such as stainless steel or reinforced plastic, the holder is designed to resist moisture and wear. The length of the holder features a telescopic design, allowing it to extend and retract. Users can adjust the holder's length by sliding it along a set of tracks, locking it in place. This adaptability accommodates a range of tumbler heights, from standard glasses to taller travel mugs. The width of the tumbler holder is designed to expand outward for easy access and storage. The pull-out mechanism is user-friendly; a gentle tug on the side panels extends the width, providing enough space for multiple tumblers. The width can be locked into place to prevent accidental retraction when in use.

The base of the holder is designed with raised edges to prevent tumblers from slipping off, ensuring stability even when extended. Rubberized grips or padding can be added to the inner surface of the holder to provide additional support and prevent sliding. The design of the tumbler holder is sleek and modern, complementing the aesthetics of contemporary RO purifiers. Available in multiple color options or finishes to match the specific model of the purifier. The entire holder can be easily adjusted without the need for tools, promoting quick transitions between different tumbler sizes. The pull-out feature allows users to access their tumblers effortlessly, making it ideal for busy kitchens or offices. The holder includes safety stops to prevent over-extension and accidental collapse, ensuring reliable performance during daily use. Materials used are non-toxic and safe for food contact, adhering to health standards.

DETAIL DESCRIPTION OF DRAWINGS

Figure 1 illustrates the RO purifier (100) with the extensible tumbler holder (101) attached to the back. The holder is depicted in its retracted state, highlighting its compact design. The telescopic design for length adjustment (101) is illustrated, showing how the holder can slide in and out. The pull-out width feature (102) is also visible, indicating the ease of access for users. The tumbler holder (102) fully extended, demonstrating its capacity to hold multiple tumblers of varying sizes.

The length is extended exactly double the size of the purifier. If the purifier's length is 46 cm then the vertical extension will be 92 cm. (101)

The width / horizontal extension will be 50 cm if the width of the purifier is 25 cm. This allows the user to keep any kind of tumbler. (102)

WE CLAIM

Claim 1: An extensible tumbler holder for a reverse osmosis (RO) purifier (100), comprising a tumbler holder tray (102) attached at the below of the RO purifier (100). An extensible arm (101) connected to the base member, and a holder mechanism (102) at the end of the extensible arm, adapted to securely accommodate tumblers of varying diameters.

Claim2: An extensible tumbler holder of claim 1, wherein the holder mechanism (102) includes a non-slip grip material to prevent tumblers from slipping.

Claim 3: An extensible tumbler holder of claim 1, wherein the holder mechanism (102) is designed to support tumblers of various shapes, including cylindrical and tapered forms.

Claim 4: An extensible tumbler holder of claim 1, wherein the extensible arm can be adjusted in length (101), allowing for customizable positioning of the tumbler.

Claim 5: An extensible tumbler holder of claim 1, wherein the extensible arm can be adjusted in width (102), allowing for customizable positioning of the tumbler



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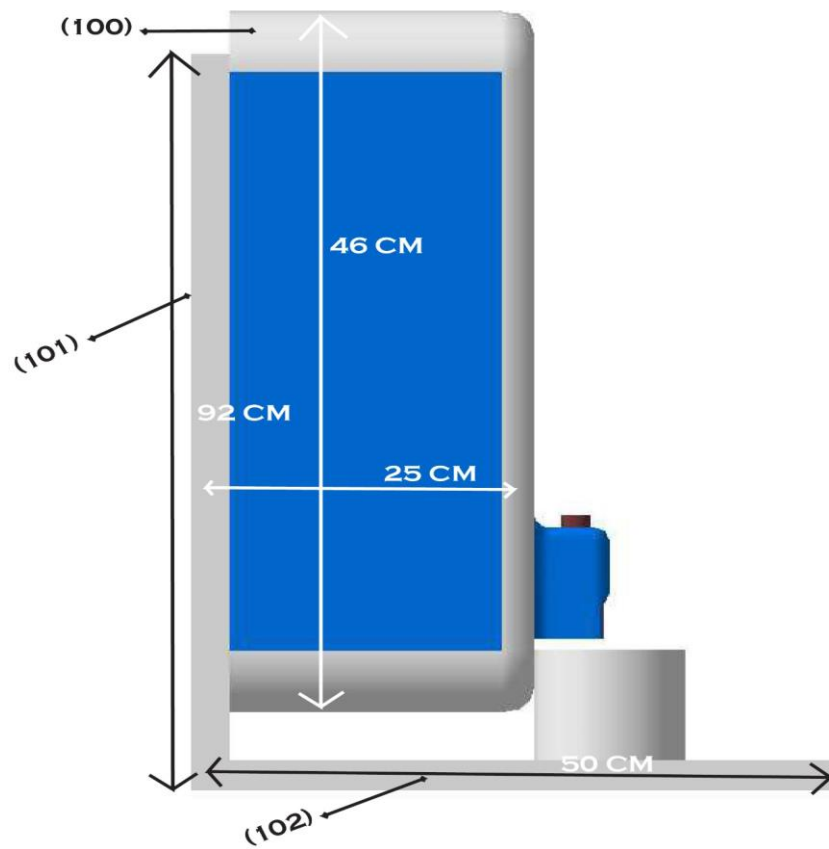


Fig 1


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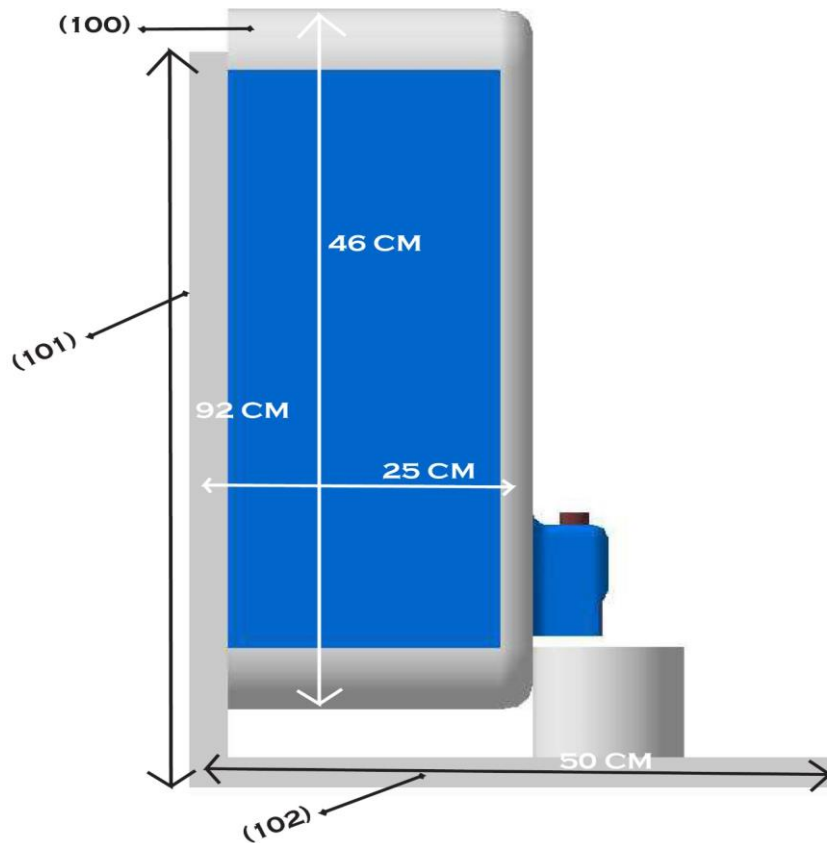
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ABSTRACT OF THE INVENTION

The Extensible Tumbler Holder for Reverse Osmosis (RO) Purifiers (100) is an innovative accessory designed to enhance user convenience and optimize space in kitchen environments. This holder (101) features a base member that securely attaches to the RO purifier (100), coupled with an adjustable extensible arm that accommodates tumblers of varying sizes and shapes. The holder mechanism (102) is constructed from durable, corrosion-resistant materials, the design emphasizes hygiene and ease of cleaning the invention is providing an ultimate solution for hydration needs in contemporary kitchens.



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